

Literature-status packet

Mean Li–Yorke chaos without distributional chaos

Status. literature__already__answered. This note records a direct, explicit resolution in a separate later paper.

Original question. N. C. Bernardes Jr., A. Bonilla, and A. Peris, *Mean Li–Yorke chaos in Banach spaces*, arXiv:1804.03900, Question 16 on PDF page 10, ask:

Is there a Banach (or Hilbert) space operator which is mean Li–Yorke chaotic but is not distributionally chaotic?

The introduction of arXiv:2511.06207 repeats this question as open.

Decisive later answer. Jian Li and Jie Li, *Power growth of mean-L-stable operators on Banach spaces*, arXiv:2608.09694, Proposition 6.1 on PDF pages 23–24, prove that their operator T_1 on X_1 is densely mean Li–Yorke chaotic but not distributionally chaotic. They prove the stronger statement that every pair in $X_1 \times X_1$ is distributionally asymptotic. Immediately after the proof they state that Proposition 6.1 answers Question 16.

Answer. The answer is affirmative on Banach spaces. The same later paper also gives the Hilbert-space contrast: Proposition 6.2, combined with its Hilbert-space mean-L-stability theorem, forces densely mean Li–Yorke chaotic Hilbert-space operators to be densely distributionally chaotic.

Provenance and scope. This packet records retrieval of a literature answer posted in August 2026; it does not claim a new proof. The identification is exact because the supporting paper cites the original Question 16 and explicitly says its Proposition 6.1 answers it.

References

- [1] N. C. Bernardes Jr., A. Bonilla, and A. Peris, *Mean Li–Yorke chaos in Banach spaces*, J. Funct. Anal. **278** (2020), 108343; arXiv:1804.03900.
- [2] J. Li and J. Li, *Power growth of mean-L-stable operators on Banach spaces*, arXiv:2608.09694, Proposition 6.1.
- [3] J. Li, X. Wang, and J. Zhao, *Mean Li–Yorke chaos for a sequence of operators on Banach spaces*, Acta Math. Sci. **46** (2026), 1187–1207; arXiv:2511.06207.